

Spatial Ergodicity for Stochastic Heat Equation

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Stochastic Heat Equation

$$\begin{cases} \left(\frac{\partial}{\partial t} - \frac{1}{2} \Delta \right) u(t, x) = \sigma(u(t, x)) \dot{W}(t, x), & t > 0, x \in \mathbb{R}^d \\ u(0, \cdot) \equiv 1 \end{cases} \quad (\text{SHE})$$

1. σ is Lipschitzian.
2. $\dot{W}$ is white in time and colored in space:

$$\mathbb{E} \left[\dot{W}(t, x) \dot{W}(s, y) \right] = \delta_0(t - s) f(x - y)$$

with f being a nonneg. and nonneg. definite measure that satisfies

$$\int_{\mathbb{R}^d} \frac{\hat{f}(dx)}{\lambda + \|x\|^2} < \infty \quad \text{for one and hence all } \lambda > 0. \quad (\text{Dalang})$$

Ergodicity of the solution

Theorem. (C., Khoshnevisan, Nualart, Pu '19+) $u(t)$ is a stationary random field for every $t > 0$. Moreover,

$$\left. \begin{array}{l} \hat{f}(\{0\}) = 0 \\ \Updownarrow \\ f([-N, N]^d) = o(N^d) \text{ as } N \rightarrow \infty \end{array} \right\} \implies u(t) \text{ is spatially ergodic}$$
$$\Updownarrow$$

$$\frac{1}{N^d} \int_{[0, N]^d} g(u(t, x)) dx = \mathbb{E} [g(u(t, 0))] \quad \forall g \in \text{Lip}.$$

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One simplest ergodic example

- ▶ Let $X_0, X_1, \dots$ are i.i.d. a stationary sequence on $(\Omega = \mathbb{R}^{\mathbb{N}}, \mathcal{F}, \mathbb{P})$

- ▶ Let φ be the shift operator on Ω , i.e., $\forall \omega = (\omega_0, \omega_1, \dots) \in \Omega$.

$$\varphi(\omega_0, \omega_1, \dots) = (\omega_1, \omega_2, \dots). \quad (\text{Measure preserving map})$$

- ▶ $A \in \mathcal{F}$ is *invariant* if $\mathbb{P}(A \Delta \varphi^{-1}(A)) = 0$. Let $\mathcal{I}$ be the invariant σ -field.

- ▶ $\forall A \in \mathcal{I}, \quad \{\omega : \omega \in A\} = \{\omega : \varphi(\omega) \in A\} \in \sigma(X_1, X_2, \dots)$

$$\Rightarrow A \in \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \dots) =: \mathcal{T} \quad (\text{the tail } \sigma\text{-field})$$

$$\Rightarrow \mathcal{I} \subseteq \mathcal{T}$$

$$\Rightarrow \mathcal{I} \text{ is trivial (by Kolmogorov's 0-1 law)}$$

- ▶ Conclusion:

The i.i.d. sequence is ergodic.

i.e., the measure-preserving map φ on the sequence space $(\Omega, \mathcal{F}, \mathbb{P})$ is ergodic.

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$$u(t, x) = 1 + \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(u(s, y)) W(ds dy)$$

- Fix arbitrary $t > 0$ and $g \in \text{Lip}$.
- $u(t)$ is stationary, i.e., the finite-dim. distribution of $u \circ \theta_y$ does not depend on $y \in \mathbb{R}^d \Rightarrow g(u(t))$ is stationary.
- *Ergodic theorem:*

$$\frac{1}{N^d} \int_{[0, N]^d} g(u(t, x)) dx = \mathbb{E} [g(u(t, 0)) | \mathcal{I}] \quad \text{a.s. \& in } L^2(\Omega)$$

- To show that $g(u(t))$ is ergodic, we need to show that the invariant σ -field $\mathcal{I}$ is trivial, that is,

$$\left(u(t, \zeta^1), \dots, u(t, \zeta^k) \right) \perp \mathcal{I}, \quad \forall k \in \mathbb{N}, \forall \zeta^1, \dots, \zeta^k \in \mathbb{R}^d,$$

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$$\mathbb{E} \left[\exp \left\{ i \sum_{j=1}^k z_j u(t, \zeta^j) \right\} \middle| \mathcal{I} \right] \stackrel{?}{=} \mathbb{E} \left[\exp \left\{ i \sum_{j=1}^k z_j u(t, \zeta^j) \right\} \right]$$

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Proof of our version of Poincaré inequality

► **Clark-Ocone formula:**

$$F = \mathbb{E}(F) + \int_0^\infty \int_{\mathbb{R}^d} \mathbb{E}[D_{s,z}F | \mathcal{F}_s] W(ds dz) \quad \text{a.s. } \forall F \in \mathbb{D}^{1,2}.$$

Poincaré inequality: $\text{Cov}(F_1, F_2) \leq \mathbb{E}(\langle DF_1, DF_2 \rangle_{\mathcal{H}}).$

- **Proposition.** (C.&Huang'18, C., Khoshnevisan, Nualart, Pu '19)
For any $(t, x) \in [0, T] \times \mathbb{R}^d$ and $k \geq 2$, we have $u(t, x) \in \mathbb{D}^{1,k}$ and

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- Set $\mathcal{G}(x) := \prod_{j=1}^k g_j(u(t, x + \zeta^j))$. We need to compute

$$\text{Var} \left(N^{-d} \int_{[0,N]^d} \mathcal{G}(x) dx \right) = N^{-2d} \iint_{[0,N]^{2d}} \text{Cov}(\mathcal{G}(x), \mathcal{G}(y)) dx dy$$

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► Notice that

$$\begin{aligned}
 & |\text{Cov}(\mathcal{G}(x), \mathcal{G}(y))| \\
 &= \left| \int_0^t ds \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} dw \mathbb{E} \{ \mathbb{E}(D_{s,z} \mathcal{G}(x) \mid \mathcal{F}_s) \cdot \mathbb{E}(D_{s,w} \mathcal{G}(y) \mid \mathcal{F}_s) \} f(z-w) \right| \\
 &\leq \int_0^t ds \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} dw \|D_{s,z} \mathcal{G}(x)\|_2 \|D_{s,w} \mathcal{G}(y)\|_2 f(z-w).
 \end{aligned}$$

► Because

$$D_{s,z} \mathcal{G}(x) = \sum_{j_0=1}^k \left(\prod_{\substack{j=1 \\ j \neq j_0}}^k g_j(u(t, x + \zeta^j)) \right) g'_{j_0}(u(t, x + \zeta^{j_0})) D_{s,z} u(t, x + \zeta^{j_0}).$$

we see that $\|D_{s,z} \mathcal{G}(x)\|_2 \leq C \sum_{j=1}^k G(t-s, x + \zeta^j - z).$

► Therefore,

$$\text{Var} \left(N^{-d} \int_{[0,N]^d} \mathcal{G}(x) dx \right) \leq \dots \leq C \frac{k^2 t}{N^d} f([-N, N]^d). \quad \square$$

- Notice that

$$\begin{aligned}
 & |\text{Cov}(\mathcal{G}(x), \mathcal{G}(y))| \\
 &= \left| \int_0^t ds \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} dw \mathbb{E} \{ \mathbb{E}(D_{s,z} \mathcal{G}(x) \mid \mathcal{F}_s) \cdot \mathbb{E}(D_{s,w} \mathcal{G}(y) \mid \mathcal{F}_s) \} f(z-w) \right| \\
 &\leq \int_0^t ds \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} dw \|D_{s,z} \mathcal{G}(x)\|_2 \|D_{s,w} \mathcal{G}(y)\|_2 f(z-w).
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Signed correlation function f

- ▶ The correlation f is assumed to be nonnegative and nonnegative definite and satisfies (Dalang).

f can be slightly more general, i.e., the nonnegativity can be dropped.

- ▶ Assume that f is of the following form

$$f = h * \tilde{h}, \quad \text{where } \tilde{h}(x) = h(-x).$$

- ▶ **Theorem.** (C., Khoshnevisan, Nualart, Pu '19+) Suppose that f takes above form with $h \in L^p_{loc}(\mathbb{R}^d)$ for some $p \geq 1$ and satisfies

$$\int_0^1 \left(\|h\|_{L^p(\mathbb{B}_r)} \|h\|_{L^q(\mathbb{B}_r^c)} + \|h\|_{L^2(\mathbb{B}_r^c)}^2 \right) \omega_d(r) dr < \infty$$

where $q = p/(p-1)$ and $\omega_d(r) = 1, r \log_+(1/r), r$ if $d = 1, 2$, resp. ≥ 3 . Then, (SHE) has a spatially stationary and ergodic random field solution.

- ▶ Poincaré inequality is still true with

$$f\left([-N, N]^d\right) \quad \text{replaced by} \quad \int_{[-N, N]^d} \left(|h| * |\tilde{h}| \right)(x) dx.$$

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Thank you!

Reference:

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Preprint at *arXiv:1907.11553*.